

Problems on range() Function

Each problem includes a sample input and expected output for better understanding.

SECTION A - BASIC

Core syntax, built-in operations, and straightforward practice.

1. Write a Python program to print numbers from 1 to 10 using range().

Input	No user input required.
Expected Output	1 2 3 4 5 6 7 8 9 10

2. Write a Python program to print all even numbers from 1 to 20 using range().

Input	No user input required.
Expected Output	2 4 6 8 10 12 14 16 18 20

3. Write a Python program to print all odd numbers from 1 to 20 using range().

Input	No user input required.
Expected Output	1 3 5 7 9 11 13 15 17 19

4. Write a Python program to print numbers from 10 to 1 in reverse order using range().

Input	No user input required.
Expected Output	10 9 8 7 6 5 4 3 2 1

5. Write a Python program to print the multiplication table of a number using range().

Input	Number = 5
Expected Output	5 x 1 = 5 5 x 2 = 10 ... 5 x 10 = 50

6. Write a Python program to find the sum of numbers from 1 to n using range().

Input	n = 5
Expected Output	Sum = 15

7. Write a Python program to find the factorial of a number using range().

Input	n = 5
Expected Output	Factorial = 120

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8. Write a Python program to print squares of numbers from 1 to 10 using range().

Input	No user input required.
Expected Output	1 4 9 16 25 36 49 64 81 100

9. Write a Python program to print cubes of numbers from 1 to 10 using range().

Input	No user input required.
Expected Output	1 8 27 64 125 216 343 512 729 1000

10. Write a Python program to count how many numbers between 1 and 100 are divisible by 7 using range().

Input	No user input required.
Expected Output	Count = 14

11. Write a Python program to print all numbers between 1 and 50 except multiples of 5 using range().

Input	No user input required.
Expected Output	1 2 3 4 6 7 8 9 11 ... 49

12. Write a Python program to print the Fibonacci series up to n terms using range().

Input	n = 7
Expected Output	0 1 1 2 3 5 8

13. Write a Python program to print the ASCII values of uppercase letters using range().

Input	No user input required.
Expected Output	A = 65 B = 66 ... Z = 90

14. Write a Python program to generate a list of numbers from 100 to 0 with a step of -10 using range().

Input	No user input required.
Expected Output	[100, 90, 80, 70, 60, 50, 40, 30, 20, 10, 0]

15. Write a Python program to print all leap years between two given years using range().

Input	Start year = 2000 End year = 2020
Expected Output	2000 2004 2008 2012 2016 2020

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16. Write a Python program to display a star pattern using nested loops and range().

Input	Rows = 4
Expected Output	<pre>* * * * * * * * * *</pre>

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SECTION B - INTERMEDIATE

Multi-step logic and practical manipulation of the topic.

1. Write a Python program to print numbers between 1 and 50 that are divisible by both 3 and 5 using range().

Input	No user input required.
Expected Output	15 30 45

2. Write a Python program to find the sum of all even numbers from 1 to n using range().

Input	n = 10
Expected Output	Sum = 30

3. Write a Python program to print all divisors of a number using range().

Input	n = 24
Expected Output	1 2 3 4 6 8 12 24

4. Write a Python program to print all prime numbers between two given numbers using range().

Input	Start = 10 End = 30
Expected Output	11 13 17 19 23 29

5. Write a Python program to print multiplication tables from 2 to 5 using nested range() loops.

Input	No user input required.
Expected Output	2 x 1 = 2 ... 2 x 10 = 20 ... 5 x 1 = 5 ... 5 x 10 = 50

6. Write a Python program to display a number triangle using range().

Input	Rows = 4
Expected Output	1 1 2 1 2 3 1 2 3 4

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7. Write a Python program to generate all coordinate pairs for rows 0-1 and columns 0-2 using nested range().

Input	No user input required.
Expected Output	(0, 0) (0, 1) (0, 2) (1, 0) (1, 1) (1, 2)

8. Write a Python program to generate the sequence 2, 5, 8, ... up to 20 using range().

Input	No user input required.
Expected Output	2 5 8 11 14 17 20

9. Write a Python program to find the sum of squares from 1 to n using range().

Input	n = 5
Expected Output	Sum of squares = 55

10. Write a Python program to display every index and value of a list using range(len(list)).

Input	List = ['A', 'B', 'C', 'D']
Expected Output	0 -> A 1 -> B 2 -> C 3 -> D

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SECTION C - ADVANCED

Algorithmic problem solving and deeper application of the topic.

1. Write a Python program to print Pascal's triangle for n rows using nested range() loops.

Input	Rows = 5
Expected Output	<pre>1 1 1 1 2 1 1 3 3 1 1 4 6 4 1</pre>

2. Write a Python program to print a diamond star pattern using range().

Input	Half-height = 4
Expected Output	<pre>* *** ***** ***** ***** *** *</pre>

3. Write a Python program to print all Armstrong numbers from 100 to 999 using range().

Input	No user input required.
Expected Output	153 370 371 407

4. Write a Python program to find all twin-prime pairs between 1 and 50 using range().

Input	No user input required.
Expected Output	(3, 5) (5, 7) (11, 13) (17, 19) (29, 31) (41, 43)

5. Write a Python program to transpose a matrix by iterating through row and column indices with range().

Input	[[1, 2, 3], [4, 5, 6]]
Expected Output	[[1, 4], [2, 5], [3, 6]]

6. Write a Python program to multiply two 2 x 2 matrices using three nested range() loops.

Input	<pre>A = [[1, 2], [3, 4]] B = [[5, 6], [7, 8]]</pre>
Expected Output	[[19, 22], [43, 50]]

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7. Write a Python program to sort a list using bubble sort implemented with nested range() loops.

Input	[5, 1, 4, 2, 8]
Expected Output	[1, 2, 4, 5, 8]

8. Write a Python program to generate all prime numbers up to n using a sieve implemented with range().

Input	n = 30
Expected Output	2 3 5 7 11 13 17 19 23 29

9. Write a Python program to display a hollow square star pattern using nested range() loops.

Input	n = 5
Expected Output	***** * * * * * * *****

10. Write a Python program to find all Pythagorean triples with values up to 20 using nested range() loops.

Input	Maximum value = 20
Expected Output	(3,4,5) (5,12,13) (6,8,10) (8,15,17) (9,12,15) (12,16,20)